

E-COMMERCE GROCERY CHATBOT MASTER TRAINING DATASET

Root-to-Top Hierarchical Intent Taxonomy, Dialogue DST & Gated Security

EXECUTIVE DATASET OVERVIEW

Developing high-accuracy, production-ready intent classification models for e-commerce grocery and produce delivery systems requires highly structured, hierarchical, domain-specific training data. In transactional systems, traditional flat models suffer from scalability issues and classification noise due to correlated labels (e.g., distinguishing basic payments from complex EBT-SNAP payment rules). This manual provides an exhaustive, root-to-top training dataset structured around a three-tier hierarchical architecture, paired with modern conversational engineering standards including Dialogue State Tracking (DST), Optical Character Recognition (OCR) multimodal integration, and Zero-Trust confidence-calibrated Handler Contracts. Use this data directly to train your classifier models (BERT, DeBERTa, or few-shot generative LLMs) and configure your transactional gates securely.

Note on Operational Standards: The dataset mappings, pricing strategies (variable weight, unit pricing), and post-purchase rules are aligned with standard digital retail policies modeled on ACME Markets Online Grocery FAQ and state-level EBT electronic payment regulations.

DOCUMENT PROFILE & QUALITY ASSURANCE CERTIFICATION

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1. ARCHITECTURAL SCHEMAS & TAXONOMY THEORY

A major failure mode in e-commerce intent routers is the **Argmax Label Trap**. Standard architectures take the highest-scoring intent prediction directly to an automated database transaction. If a customer says *'Cancel my order'*, a naive router with 0.85 confidence might trigger cancellation, without verifying whether the order is unfulfilled, whether it was placed today, or which card was charged. As outlined in conversational research, intent classification must be treated as a **routing proposal**, not a verdict. Security, compliance, and user safety are enforced by a downstream **Gated Handler Contract**.

The Three-Tier Root-to-Top Taxonomy

To solve the scalability issues associated with flat 100+ intent sets, modern classifiers use a hierarchical taxonomy. By partitioning labels into nested classifiers, each individual classification node handles fewer than 50 categories. This drastically minimizes model latency, enables parameter efficiency, and improves overall F1-scores.

Taxonomy Structure:

Level 1 (Root Nodes)	Level 2 (Sub-categories)	Level 3 (Leaf Intents)
Product Discovery	Deli, Bakery & Fresh Produce Meal Planning & Recipes Dietary Preferences	search_fresh_produce inquire_variable_weight_pricing build_basket_from_recipe
Order Management	Scheduling & Delivery Slots Order Adjustments	reserve_delivery_slot modify_unfulfilled_order
Billing & Payments	Traditional Gateways Government & HSA Benefits Temporary Charges	pay_with_ebt_snap pay_with_healthcare_benefit explain_preauthorization_hold
Post-Purchase CX	Satisfaction Guarantees Category Restrictions	report_missing_grocery_item reject_substituted_item

Key Axiom of Gated Safety: High-stakes transactions (e.g., EBT-SNAP processing, unfulfilled order modification, refund allocations) must have individual, custom-calibrated confidence bars and mandatory validation checks. A balance query can run at 0.65 confidence, but an EBT transaction or order cancellation must demand at least 0.85-0.90 confidence paired with a mandatory confirmation screen.

2. PRODUCT DISCOVERY & CART BUILDING

This node is designed to handle the complex, conversational catalog querying phase. In grocery e-commerce, product discovery is uniquely complicated by **variable weight pricing** (e.g., ordering Bosc pears by the piece vs. by the pound) and **recipe-to-cart parsing**. Below is the production-grade dataset for Level 3 leaf intents in this category.

Leaf Intent / Tag	Required Slots	Production-Grade Training Patterns (Customer Queries)
search_fresh_produce	produce_type quantity ripeness (opt)	- Add 5 ripe avocados and three green bananas to my shopping cart - Do you have organic honeycrisp apples in stock today? - Find some fresh strawberries and add 2 packs - Search for organic baby spinach 10oz - Put four yellow peaches in my basket - I need some seedless watermelons, is there any fresh ones?
inquire_variable_weight_pricing	product_name unit_type	- If I buy 3 apples, am I paying by the piece or the weight? - I want to order sliced turkey breast from the deli counter, how do I specify weight? - Are watermelons priced per pound or per unit? - How is the beef chuck roast priced? - If I select bananas, does the shopper weigh them? - Tell me how variable weight items are charged at checkout.
build_basket_from_recipe	recipe_name servings exclude_allergen	- I want to make lasagne on Monday and vegetarian patties on Tuesday. Please add the ingredients to my basket - Find a recipe for gluten-free pizza for 4 people and put all of it in my cart - Put ingredients for chocolate chip cookies for 10 servings in my bag - Add all the stuff needed to cook beef stroganoff tonight - What do I need to buy to make fresh guacamole? Put it in my cart.
filter_dietary_restrictions	diet_type allergen (opt)	- Filter my catalog search to only show Keto-friendly items - Show me dairy-free ice cream brands in your store - I am highly allergic to peanuts, hide all peanut-containing snacks - Are there any vegan and gluten-free bread options available? - Filter out anything that contains soy or gluten - Only show organic, non-GMO produce options.

Contextual Response Mappings

`search\_fresh\_produce` response logic: Trigger elasticsearch on catalog with parsed slots. If item is in-stock, return product list with direct 'Add to Cart' links. If item is out-of-stock, fallback to substitution preferences.

`inquire\_variable\_weight\_pricing` response logic: Explain clearly: *'For items sold by weight (such as produce and deli meats), the checkout total uses an Estimated Price based on average weight. The final exact price is determined on the scale when your shopper select and weighs your exact items.'*

3. ORDER MANAGEMENT & SCHEDULE BOOKING

Grocery delivery scheduling relies heavily on localized zip code checks and tight unfulfilled order rules. According to operational policy (e.g., ACME Markets), customers can only modify items or change delivery/pickup times **on the same day the order is placed**. Once fulfillment begins, the slot is locked.

Leaf Intent / Tag	Required Slots	Production-Grade Training Patterns (Customer Queries)
reserve_delivery_slot	zipcode delivery_time	- Can I schedule a delivery to 19104 tomorrow morning? - Reserve a delivery window between 2 PM and 4 PM for my address - I want to book a delivery slot for zipcode 08003 - Is there any local delivery slots open tonight before 9 PM? - Book a home delivery window for this Saturday - What times are available for grocery delivery in my neighborhood?
reserve_driveup_pickup_slot	store_location pickup_time	- Can I schedule a DriveUp curbside pickup at the Main Street store for tomorrow at 3 PM? - Reserve a pickup slot for tonight at the cherry hill location - Is there a pickup slot open tonight before 8 PM at store #445? - Schedule curbside pickup for this afternoon - Reserve a DriveUp & Go time slot for tomorrow morning - I need to book a curbside spot for 5 PM today.
modify_unfulfilled_order	order_id modification_action item_name (opt)	- Please edit my order #88271 to add a carton of milk - I placed an order an hour ago, can I change the pickup time to 6 PM? - Add two more bags of chips to my active grocery order - I need to swap the whole milk in my order for 2% milk - Can I add organic apples to my pending delivery scheduled for tonight? - Remove the ribeye steaks from my order #55491.
cancel_unfulfilled_order	order_id	- Cancel my grocery order #44591 immediately - I want to cancel the delivery I scheduled for tomorrow - Please void my pickup order placed this morning - How do I terminate my active unfulfilled order? - Please cancel order #10023 and refund my card - I made a mistake, please cancel my scheduled order right now.

Fulfillment Schedule Constraints & Thresholds

Standard delivery and pickup slots are bound to **8:00 AM – 9:00 PM local time**. The reservation is held in cache for up to **90 minutes** while the customer shops. If checkout is not completed within 90 minutes, the slot is released to avoid operational bottlenecks.

4. BILLING, PAYMENTS & WELFARE BENEFITS

Grocery e-commerce operates within stringent regulatory boundaries. In addition to credit cards, systems must handle **EBT-SNAP (Supplemental Nutrition Assistance Program)**, EBT-Cash (TANF benefits), and specialized healthcare benefits (FSA/HSA) online. Temporary pre-authorization holds must also be explained to prevent charge disputes.

Leaf Intent / Tag	Required Slots	Production-Grade Training Patterns (Customer Queries)
pay_with_ebt_snap	benefit_program card_number (opt)	- How do I use my SNAP EBT card online to buy fresh fruits? - Can I enter my EBT card number to pay for EBT eligible items? - Does your checkout accept electronic food stamps online? - I want to pay for my organic vegetables with my SNAP card - How do I link my EBT card to my digital wallet for payment? - Can I use my SNAP benefits for curbside pickup orders?
pay_with_healthcare_benefit	benefit_program	- Can I use my FSA or HSA card to pay for vitamins online? - Do you accept healthcare benefit cards for approved grocery items? - Can I pay with my HSA card for organic baby formula? - Is my insurance benefit card accepted at online checkout? - How do I find out which items in my cart are FSA/HSA eligible? - Can I add my private insurance benefit card to my account?
explain_preauthorization_hold	payment_method	- Why is there a pending charge on my credit card that is higher than my bill? - Why did you hold extra money on my debit card? - How long does it take for the estimated total hold to be released? - Explain the pre-authorization hold on my PayPal account - My order was \$45 but my bank statement shows a hold for \$60. Why? - When will the temporary hold from my cancelled order be returned?

Welfare Benefits Rules & Hold Durations

EBT-SNAP Online Compliance: SNAP benefits can ONLY be applied to SNAP-eligible food items (fruits, vegetables, meats, dairy, bread). Delivery fees, service fees, bag fees, and non-eligible items (alcohol, pet food, household cleaners) must be paid with a secondary traditional credit/debit card at checkout.

Pre-Authorization Hold Lifespan: Pre-authorized estimated totals are held by the bank on the day the order is placed to handle potential substitutions and weighed items. The bank typically takes **3 to 5 business days** after delivery or pickup to release the hold and post the actual final charge.

5. POST-PURCHASE CX & QUALITY GUARANTEES

Post-purchase intents have high financial blast radiuses and strict time restrictions. Under the **100% Item Satisfaction Guarantee**, missing items, damaged produce, or rejected substitutions must be reported within **48 hours** to trigger automated refunds or store credits.

Leaf Intent / Tag	Required Slots	Production-Grade Training Patterns (Customer Queries)
report_missing_grocery_item	order_id item_name	- My grocery delivery just arrived but the ribeye steak is missing - Order #55492 is missing the bagels and organic butter - I was charged for some fresh strawberries but they weren't in the bags - My delivery is missing three items from the receipt - How do I report that my milk carton was left out of my order? - I didn't receive the orange juice I paid for in my pickup order
reject_substituted_item	order_id item_name substituted_item	- The shopper replaced my organic eggs with regular eggs, I want a refund - The substitute cheesecake is damaged, I'd like a credit - I want to reject the substitution of store-brand bread for Sara Lee - The shopper substituted regular milk but I wanted soy milk, refund me - I did not agree to the brand swap on my paper towels - The replaced item is not acceptable, please refund my EBT card
check_restricted_category_policy	product_category order_id (opt)	- Can I return the baby formula I bought this morning? - I want to refund the wine I received in my delivery - What is your return policy on tobacco products and cigarettes? - Can I return alcohol bought online back to the physical store? - I ordered baby formula by mistake, how do I get a credit? - Are there any restrictions on returning diapers or baby food?

Operational Refund Exclusions & Time Limits

The 48-Hour Hard Rule: Missing items and damaged goods must be reported within 48 hours. Any requests submitted past this window are automatically routed to human customer support agents for manual audit.

Federal Exclusions: Baby formula, alcoholic beverages, and tobacco products cannot be returned, refunded, or credited once they are accepted at delivery or picked up at curbside due to federal health and licensing regulations.

6. DIALOGUE FLOWS & DIALOGUE STATE TRACKING

Below are three realistic, multi-turn dialogue transcripts showing how intent classification, slot extraction, and state tracking (DST) function in real-world scenarios. Slot variables are highlighted in bold.

Scenario 1: Fresh Produce Search & Ripeness Slot Filling

User: I need some fresh avocados and bananas sent to my house.

[State: Intent proposal = `search\_fresh\_produce`, Slots: produce_type=["avocado", "banana"]]

Bot (Flippi Followup): I can help with that! How many of each would you like? Also, do you have any ripeness preferences (for example, green or ready-to-eat)?

User: Get me **5 ripe** avocados and **three green** bananas please.

[State: Intent proposed = `search\_fresh\_produce` (confidence 0.98), Slots: produce_type=["avocado", "banana"], quantity=[5, 3], ripeness=["ripe", "green"]]

Bot: Perfect! I've added 5 ripe avocados and 3 green bananas to your cart.

Scenario 2: Variable Weight pricing Explanation

User: I want to order some sliced turkey from the deli counter, but it says \$8.99 per pound. How do I choose how much I get? and how will I be charged?

[State: Intent proposal = `inquire\_variable\_weight\_pricing`, Slots: product_name="sliced turkey", unit_type="pound"]

Bot: Great question! For deli items like **sliced turkey**, you can select the weight you want in fractions of a pound (like 0.5 lb or 1 lb) when adding it to your cart.

At checkout, your bill will show an Estimated Total. When your Personal Shopper slices the turkey at the counter, they will weigh it on the scale. Your final receipt will reflect the exact weight of the turkey you receive.

Scenario 3: EBT SNAP Payment Online Gate

User: Can I use my food stamp EBT card online to buy groceries?

[State: Intent proposed = `pay\_with\_ebt\_snap` (confidence 0.97), Slots: benefit_program="EBT SNAP"]

Bot: Yes, we absolutely accept **EBT SNAP** online for pickup and delivery! You can buy fresh fruits, vegetables, meat, dairy, and bread with your benefits.

Please note that delivery fees, bag fees, and household cleaners are not eligible under SNAP regulations and must be paid with a backup credit or debit card at checkout. Would you like me to walk you through linking your EBT card?

7. SECURING INTERACTIONS: THE HANDLER CONTRACT

To guarantee absolute safety in your transactional e-commerce chatbot, implement the **Handler Contract** system (as specified in modern conversational engineering). Below is the clean, executable Python class mapping intents to safety gates.

```
from dataclasses import dataclass
from typing import Callable, Dict, Any, Tuple

@dataclass(frozen=True)
class HandlerContract:
    handler: Callable
    min_confidence: float          # calibrated threshold THIS action requires
    required_slots: Tuple[str, ...] # entities the extractor must have produced
    required_auth: str             # "none" | "session" | "otp"
    confirm: bool = False          # high blast radius: confirm parsed action

# Define your custom-calibrated contracts based on action risk
CONTRACTS = {
    "search_fresh_produce": HandlerContract(
        handler=execute_catalog_search,
        min_confidence=0.60,          # Low risk: easy to autocorrect
        required_slots=("produce_type",),
        required_auth="none"
    ),
    "modify_unfulfilled_order": HandlerContract(
        handler=execute_order_edit,
        min_confidence=0.85,          # High risk: modifying active order
        required_slots=("order_id", "modification_action"),
        required_auth="session",
        confirm=True                  # Mandatory confirmation screen
    ),
    "pay_with_ebt_snap": HandlerContract(
        handler=trigger_ebt_wallet_link,
        min_confidence=0.90,          # Financial compliance must have high confidence
        required_slots=(),
        required_auth="otp",          # Force SMS step-up authentication
        confirm=True
    ),
    "reject_substituted_item": HandlerContract(
        handler=allocate_substitution_refund,
        min_confidence=0.85,          # Money movement requires high assurance
        required_slots=("order_id", "item_name"),
        required_auth="session",
        confirm=True
    )
}

def route_safely(prediction: Dict[str, Any], entities: Dict[str, Any], ctx: Any) -> Any:
    """
    Gateway router enforcing the contract.
    Returns: execution result, or a safe fallback action (clarify, slot fill, confirm).
    """
    intent = prediction.get("intent")
    confidence = prediction.get("confidence", 0.0)
    contract = CONTRACTS.get(intent)

    if contract is None:
        return trigger_agent_handoff("I could not verify that request safely.")

    # 1. Confidence Gate
```

```
if confidence < contract.min_confidence:
    return trigger_intent_clarification(intent)

# 2. Slot Validation Gate (Presence and validation of required variables)
missing = [slot for slot in contract.required_slots if slot not in entities or not entities[slot]]
if missing:
    return trigger_slot_filling_module(missing)

# 3. Confirmation Gate (Catching confident but incorrect labels)
if contract.confirm and not ctx.user_has_confirmed:
    return trigger_user_confirmation_screen(intent, entities)

# 4. Authentication Step-Up Gate
if not user_meets_auth_level(ctx.auth_level, contract.required_auth):
    return trigger_step_up_authentication(contract.required_auth)

# All gates cleared: Proceed to run safe transaction
return contract.handler(ctx, **{s: entities[s] for s in contract.required_slots})
```

Benefits of Gated Pipeline Integration

By moving security boundaries to a single, deterministic gate layer, your business logic remains decoupled from the classifier model. Developers can change, retrain, or substitute the underlying classifier (e.g., swapping a DeBERTa model for a fine-tuned Qwen 2.5) without risking payment compliance or code breaks.

8. MULTIMODAL OCR-BERT & VISUAL RESOLUTION

In real-world grocery customer service, up to **30% of user queries contain image uploads** alongside text (photos of damaged fruits, screenshots of unapplied promo codes, or grocery receipts). Purely text-based classifiers completely fail to understand these context-rich interactions. As analyzed in the Multimodal Conversational Intent Classification (MCIC) framework, integrating image categories and **Optical Character Recognition (OCR)** is necessary to achieve high after-sales accuracy.

Multimodal Dual-Modality Fusion Architectures

To resolve multimodal chats, your backend system should run a dual-branch fusion network:

1. OCR-BERT Branch (The Textual Bridge):

An OCR model (e.g., EAST text detector + RCNN sequence recognizer) extracts characters from screenshots (such as *'Coupon expired'* or *'Minimum order not met'*). This OCR text is appended directly to the user utterance tokens split by a special `[SEP]` marker. This is highly effective because text inside images carries direct, explicit intent cues.

2. VisualBERT Branch (The Vision Bridge):

When a user uploads a photo of a damaged item (such as a bruised apple or split milk container), no text is present in the image. VisualBERT extracts local features using a pretrained ResNet-50 CNN (pooling visual feature layer with 2048 dimensions). These regional features are transformed into visual tokens, which interact directly with user text tokens in multi-layer bidirectional self-attention transformers.

Operational Guidelines for Visual Claims Verification

When a damage or quality refund claim is filed with an image, classify the image category into one of these types: ``app_screenshot`` vs. ``commodity_photo``. If classified as a ``commodity_photo`` and visual feature confidence for damage is over 0.82, automate the 100% Item Satisfaction Guarantee refund. If it is classified as ``app_screenshot`` with OCR text indicating a billing error, immediately trigger the account credit pipeline.